MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

PHYS-111 Chapter 1 Tutorial Questions

1. The density of gold is 19.3 g/cm^3 . What is this value in kilograms per cubic meter?
2. A solid piece of lead has a mass of 23.94 g and a volume of 2.10 cm^3. From these data, calculate the density of lead in SI units.
3. How many years is 1.00 gigasecond? (Assume a 365-day year.)
4. Write the following quantities using unit prefixes.

   (a) 0.005 m (b) 5000 g (c) 0.0000075 s (d) 0.00000000872 m/s
5. Express the following numbers using scientific notation.

   (a) 120000000 (b) 0.00000000034.
6. How many significant figures are in the following numbers?

   (a) 78.9$\pm$0.2 (b) 3.788×10^9 (c) 2.46×10^{-6} (d) 0.0053.
7. Which of the following equations are dimensionally correct?

   (a) $v_f = v_0 + ax$, where v_f and v_0 are final & initial velocities respectively, a is the acceleration and x is the displacement.

   (b) $y = 2\text{m} \cos(\text{kx})$, where $\text{k} = 2\ \text{m}^{-1}$, x and y are displacements in x- and y-axes respectively.
8. The position of a particle moving under uniform acceleration is some function of time and the acceleration. Find the simplest equation that describes the position of that particle.
9. What order of magnitude are the following numbers?

   (a) 0.008 6 (b). 0.002 1. (c). 720
10. (a). The position of a particle in space is described by (3.00, 4.00) m. Find the polar coordinates of this particle.

    (b). The polar coordinates of a point in space are given by (5.00 m, 37^0). Express these coordinates in rectangular form.